

Module 6: Network Security, Maintenance & Troubleshooting Procedures

Section 1: Multiple Choice

1. What is the primary purpose of a firewall in a network security infrastructure?

Answer: b) Filtering and controlling network traffic

2. What type of attack involves flooding a network with excessive traffic to disrupt normal operation?

Answer: a) Denial of Service

3. Which encryption protocol is commonly used to secure wireless network communications?

Answer: b) WPA

4. What is the purpose of a VPN (Virtual Private Network) in a network security context?

Answer: a) Encrypting network traffic to prevent eavesdropping

Section 2: True or False

1. Patch management is the process of regularly updating software and firmware to address security vulnerabilities and improve system performance.

Answer: True

2. A network administrator should perform regular backups of critical data to prevent data loss in the event of hardware failures, disasters, or security breaches.

Answer: True

3. Traceroute is a network diagnostic tool used to identify the route and measure the latency of data packets between a source and destination device.

Answer: True

Section 3: Short Answer

8. Describe the steps involved in conducting a network vulnerability assessment.

Answer:

A network vulnerability assessment is performed to identify security weaknesses in a network. Steps include:

1. Define Scope – Decide which systems, devices, and applications to scan.
2. Gather Information – Collect network details such as IP ranges, OS versions, and services running.

3. Identify Vulnerabilities – Use automated tools to scan for weak points.
4. Analyze Risks – Prioritize vulnerabilities based on severity, exploitability, and impact.
5. Report Findings – Document detected vulnerabilities with details and recommendations.
6. Fix Issues – Apply patches, update systems, close unused ports, and improve configurations.
7. Re-scan – Verify that all vulnerabilities have been fixed.

Section 4: Practical Application

9. Demonstrate how to troubleshoot network connectivity issues using the ping command.

Answer:

Steps to troubleshoot using ping:

1. Ping localhost

```
ping 127.0.0.1
```

Confirms network adapter is working.

2. Ping your own IP address

```
ipconfig  
ping <your_IP>
```

Checks internal network configuration.

3. Ping the default gateway

```
ping <gateway_IP>
```

Tests connection with the router.

4. Ping an external server

ping google.com

Verifies internet access.

5. Interpret results:

- a. Reply received: Device reachable
- b. Request timed out: Device not reachable
- c. High latency: Network congestion
- d. Packet loss: Poor connection or hardware problem

Section 5: Essay

10. Discuss the importance of regular network maintenance and the key tasks involved in maintaining network infrastructure.

Answer:

Regular network maintenance is essential to keep the network secure, stable, and efficient. Without routine maintenance, security vulnerabilities, slow performance, and unexpected outages can occur.

Importance of network maintenance:

- Improves security by identifying vulnerabilities and applying patches.
- Prevents downtime and ensures smooth business operations.
- Optimizes performance by tuning devices and reducing congestion.
- Extends hardware lifespan through monitoring and cleaning.
- Ensures compliance with organizational or legal standards.

Key maintenance tasks include:

1. Updating firmware and software to fix security flaws.
2. Monitoring network traffic to detect unusual behavior.

3. Checking hardware health (switches, routers, cables).
4. Backing up configurations and data regularly.
5. Reviewing firewall and security policies for improvements.
6. Removing unused applications, services, or user accounts.
7. Testing connectivity using tools like ping, traceroute, and network analyzers.

Regular maintenance ensures the network remains secure, fast, and reliable.